TASK - 3

What is Data Analysis?

Data analysis is the process of collecting, organizing, and studying data to find useful information and make better decisions.

It is important because it helps:

- Understand patterns and trends
- Make business decisions
- Improve products and services
- Solve real-world problems using data

Examples of data analysis:

- Sales reports
- Student result analysis
- Weather forecasting
- Social media insights

Pandas Library

Pandas is a popular Python library used for data analysis and data handling.

It works similar to Excel because it allows us to:

- Read data from files
- Organize data into tables
- Filter and sort information
- Perform calculations easily

Before using Pandas, it must be imported:

import pandas as pd

Reading Data from CSV Files

CSV (Comma Separated Values) files are used to store table-like data, similar to Excel sheets.

Pandas can read CSV files using:

import pandas as pd

```
data = pd.read_csv("students.csv")
print(data)
```

This loads the file into a table format called a DataFrame.

Exploring Data

After loading data, we can explore it to understand the dataset better.

Useful functions:

```
print(data.head())
```

Shows the first 5 rows.

```
print(data.info())
```

Displays information about columns and data types.

```
print(data.shape)
```

Shows number of rows and columns.

Exploring data helps us understand:

- Total records
- Missing values
- Data types
- Overall structure

Data Cleaning

Data cleaning means fixing incorrect, empty, or missing values in the dataset.

Common cleaning tasks:

- Removing duplicates
- Filling missing values
- Correcting wrong data
- Removing unnecessary columns

Example:

```
data = data.dropna()
```

Removes rows with missing values.

data = data.fillna(0)

Replaces missing values with `0`.

Data cleaning improves accuracy and reliability of analysis.

Simple Statistics

Pandas can calculate basic statistics easily.

Example:

print(data["Marks"].mean())

Finds average value.

print(data["Marks"].max())

Finds maximum value.

print(data["Marks"].min())

Finds minimum value.

These statistics help summarize and understand data quickly.

```
PS C:\Users\91707\Desktop\personal intro project\Week 3> python -u "c:\Users\91707\Desktop\personal intro project\Week 3\sales analysis.py"
First 5 Rows of Dataset:
         Date     Product  Quantity  Price Customer_ID Region  Total_Sales
0  01-01-2024       Phone         7  37300     CUST001   East       261100
1  02-01-2024  Headphones         4  15406     CUST002  North        61624
2  03-01-2024       Phone         2  21746     CUST003   West        43492
3  04-01-2024  Headphones         1  30895     CUST004   East        30895
4  05-01-2024      Laptop         8  39835     CUST005  North       318680

Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   Date         100 non-null    object
 1   Product      100 non-null    object
 2   Quantity     100 non-null    int64
 3   Price        100 non-null    int64
 4   Customer_ID  100 non-null    object
 5   Region       100 non-null    object
 6   Total_Sales  100 non-null    int64
dtypes: int64(3), object(4)
memory usage: 5.6+ KB
None

Missing Values:
Date           0
Product        0
Quantity       0
Price          0
Customer_ID    0
Region         0
Total_Sales    0
dtype: int64

========== SALES REPORT ==========
Total Revenue       : ₹12,365,048.00
Best Selling Product: Laptop
Highest Sale        : ₹373,932.00
Average Sales       : ₹123,650.48
==================================
PS C:\Users\91707\Desktop\personal intro project\Week 3>
```